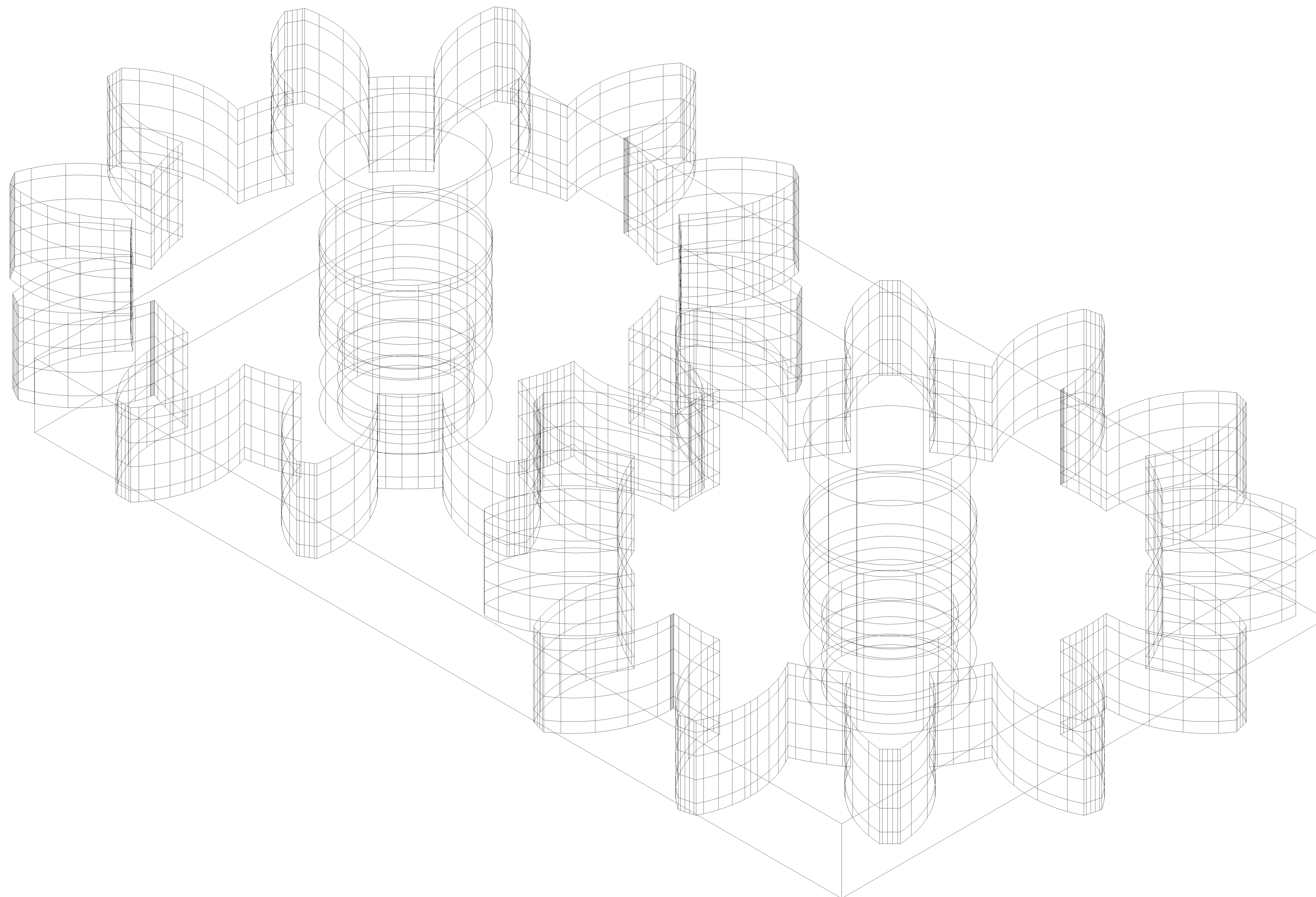
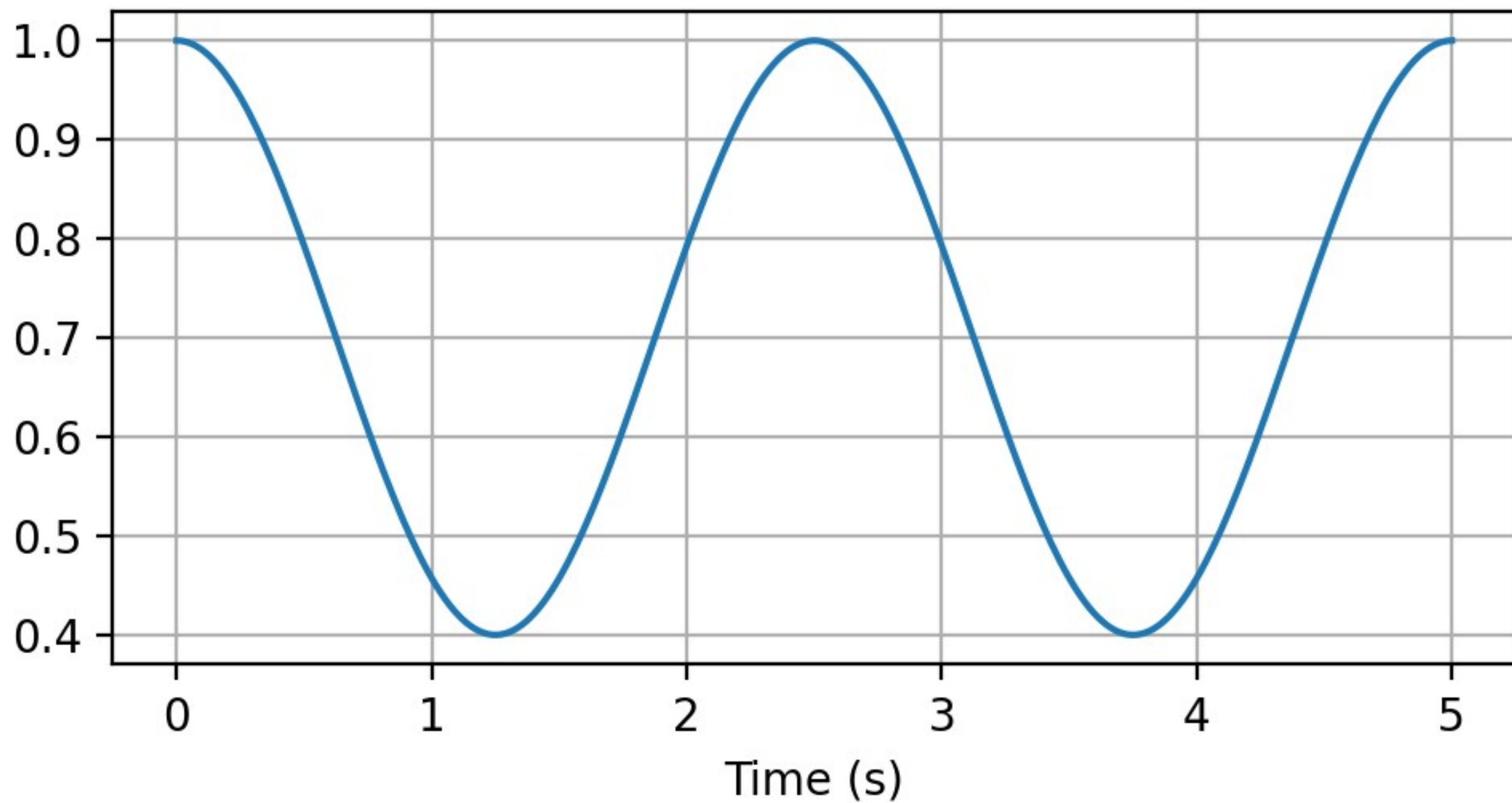
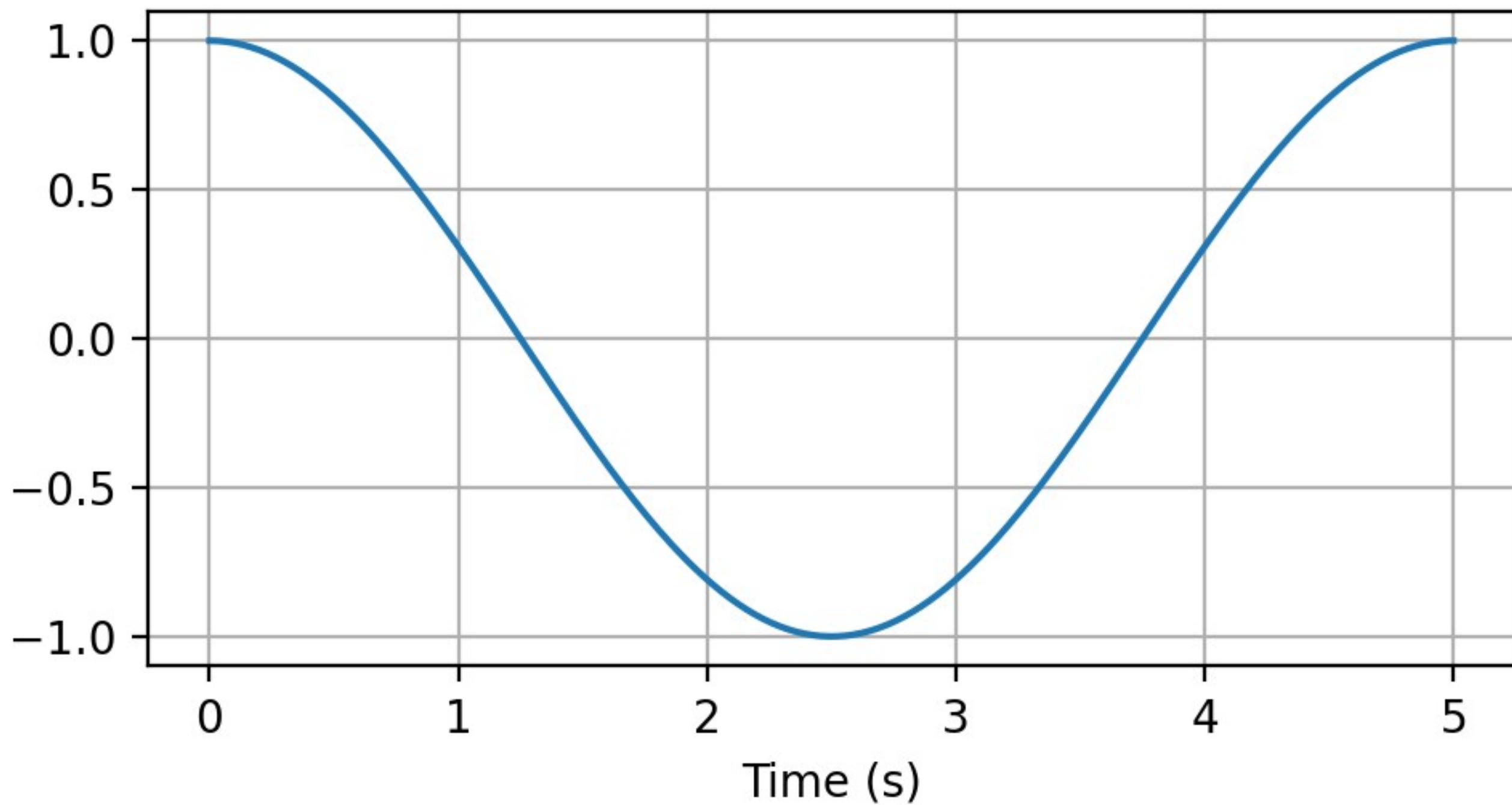




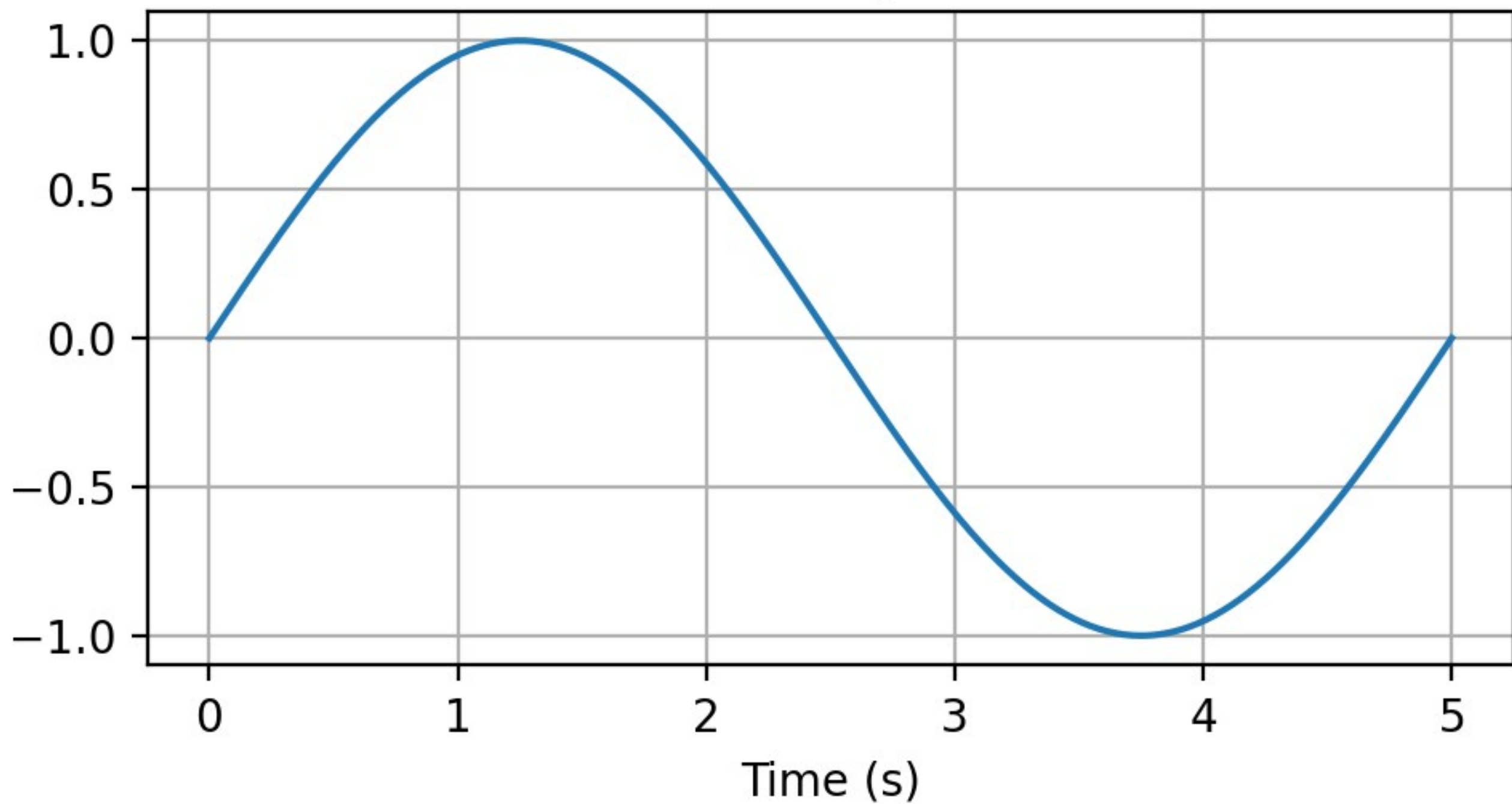
Slider Force vs Time



Slider Acceleration vs Time



Slider Velocity vs Time



Motion Simulation Report – Gear Train Mechanism

Objective: Create a spur gear train and analyze velocity, acceleration and forces using motion simulation.

Software Used: SolidWorks Motion Study (or equivalent CAD software).

Mechanism: Driver gear: 20 teeth, Driven gear: 40 teeth, Gear ratio 2:1, Input speed 120 rpm.

Procedure: Model gears, apply gear mate, assign rotary motor, run 5 s simulation, record velocity, acceleration and contact force, generate graphs and animation.

Results:

- Driver speed: 120 rpm
- Driven speed: 60 rpm (opposite direction)
- Acceleration stabilizes near zero after startup.
- Contact force remains nearly constant during steady motion.

Observation: Gear ratio maintained correctly and smooth power transmission observed.

Conclusion: The motion simulation verified correct gear train operation and produced results consistent with theoretical expectations.

References: SolidWorks Motion Guide; Khurmi & Gupta – Theory of Machines; Norton – Design of Machinery.